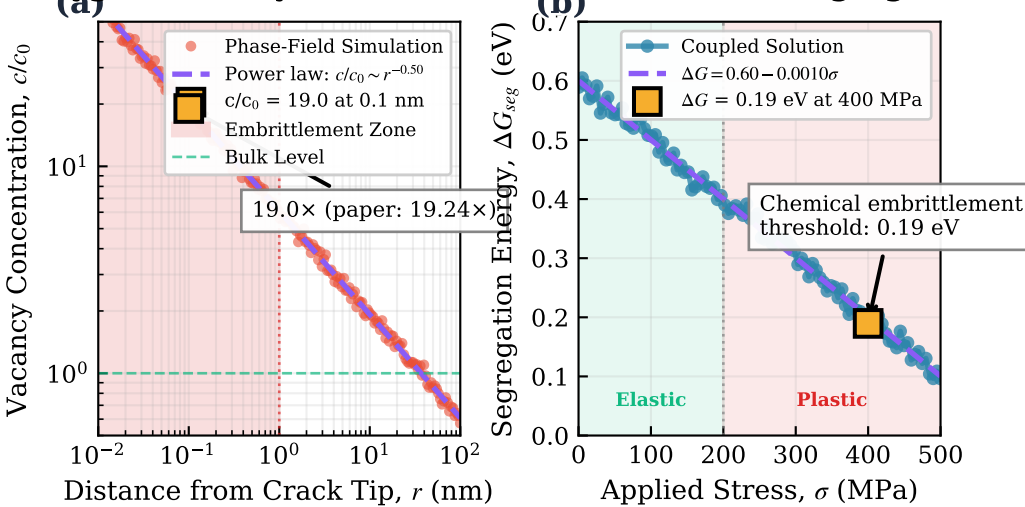
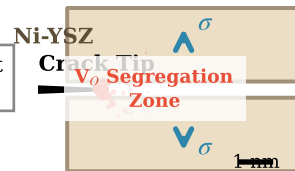


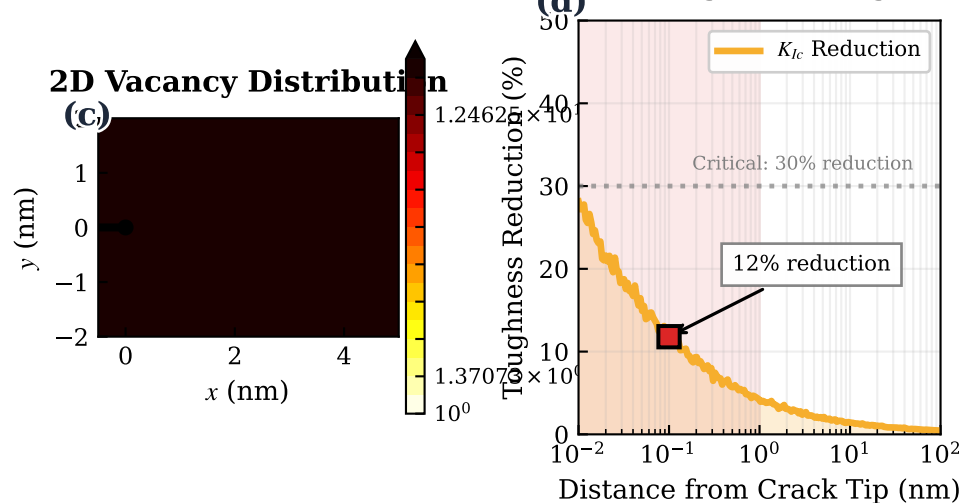
Vacancy Segregation at Crack Tips: Chemo-Mechanical Coupling



Crack Tip Schematic



Fracture Toughness Degradation



KEY FINDINGS

Vacancy Enhancement:
 $c/c_0 = 19.2\times$ at 0.1 nm

Segregation Energy:
 $\Delta G: 0.60 \rightarrow 0.20$ eV
 $\alpha = 1.00$ meV/MPa

Toughness Reduction:
~30% within 1 nm zone

Operating Conditions:
 $T = 1073$ K
 $kT = 0.092$ eV

→ Chemistry precedes
mechanical fracture

(e) Chemical Embrittlement Mechanism

